Week 4 - Vertical Profiles: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS**: selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY**: expected independent practice after the session.
- **EXTRA CHALLENGE**: optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Unless stated otherwise, use $g = 9.81\ \mathrm{m\,s^{-2}}$, $R_d = 287\ \mathrm{J\,kg^{-1}\,K^{-1}}$ and a representative pressure scale height $H = 8.2$ km.

IN CLASS Question 1: Deriving hydrostatic balance

Consider a horizontal slab of air of area A, thickness dz and density ρ.

1. Draw and label the vertical pressure forces and the weight of the slab.
2. Derive the hydrostatic equation
$$\frac{dp}{dz} = -\rho g.$$
3. State clearly what physical balance the equation represents.

EXTRA CHALLENGE Question 2: Pressure levels and atmospheric mass

At sea level take $p_s = 1013$ hPa. Under hydrostatic balance, pressure at a level is proportional to the mass of atmosphere above that level per unit area.

1. What fraction of the total atmospheric mass lies above the 850 hPa surface?
2. Repeat for 350 hPa and 10 hPa.
3. Explain why these results are useful for interpreting pressure coordinates in meteorology.

YOU SHOULD TRY Question 3: How high is the 500 hPa level?

Assume
$$p = p_0 \exp(-z/H),$$
with $p_0 = 1013$ hPa and $H = 8.2$ km. Estimate the altitude of the 500 hPa surface.

EXTRA CHALLENGE Question 4: Density with height

Assume $\rho_0 = 1.225\ \mathrm{kg\,m^{-3}}$ and the same scale height $H = 8.2$ km. Calculate the density at 5 km and 10 km using
$$\rho = \rho_0 \exp(-z/H).$$

YOU SHOULD TRY Question 5: Aircraft altimetry and pressure errors

An aircraft is actually at 9.0 km in an atmosphere described by $p = p_0 e^{-z/H}$ with $p_0 = 1013$ hPa and $H = 8.2$ km.

1. Estimate the ambient pressure.
2. Two aircraft are intended to remain vertically separated by 300 m. Approximately what pressure difference at 9 km corresponds to a 300 m altitude difference in this model?
3. Why does this illustrate the importance of pressure calibration in barometric altimetry?

YOU SHOULD TRY Question 6: Reducing a hill-station pressure to sea level

A barometer reads 950 hPa at a station 500 m above sea level. Assume an isothermal layer at 273 K.

1. Calculate the scale height $H = R_d T/g$.
2. Estimate the equivalent sea-level pressure using $p(z) = p_0 e^{-z/H}$.
3. Name two reasons why a real meteorological sea-level pressure reduction requires more care than this calculation.

IN CLASS Question 7: Lapse rate and dry stability

A sounding gives the following temperatures in the lowest 3 km:

Height (km)	0	1.0	2.0	3.0
Temperature (°C)	18	12	7	2

1. Calculate the mean environmental lapse rate over 0–3 km.
2. Compare it with the dry adiabatic lapse rate, $\Gamma_d \approx 9.8\ \mathrm{K\,km^{-1}}$.
3. Would an unsaturated parcel displaced upward through this layer tend to become warmer or colder than its environment? What does that imply for dry static stability?

EXTRA CHALLENGE Question 8: Scale height on Earth and Mars

Use

$$H = \frac{R^* T}{Mg}$$

with $R^* = 8.314\ \mathrm{J\,mol^{-1}\,K^{-1}}$.

1. For Earth use $T = 288$ K, $M = 28.97 \times 10^{-3}\ \mathrm{kg\,mol^{-1}}$, $g = 9.81\ \mathrm{m\,s^{-2}}$.
2. For Mars use $T = 210$ K, $M = 44.01 \times 10^{-3}\ \mathrm{kg\,mol^{-1}}$, $g = 3.71\ \mathrm{m\,s^{-2}}$.

Calculate both scale heights and explain why Mars can have a larger scale height despite its colder atmosphere.

EXTRA CHALLENGE Question 9: Hydrostatic pressure in an atmosphere with a constant lapse rate

Assume hydrostatic balance, ideal-gas air, and a temperature profile

$$T(z) = T_0 - \Gamma z,$$

where $T_0 = 288$ K and $\Gamma = 6.5\ \mathrm{K\,km^{-1}}$. Starting from

$$\frac{dp}{dz} = -\rho g, \qquad p = \rho R_d T,$$

show that

$$p(z) = p_0 \left(\frac{T(z)}{T_0}\right)^{g/(R_d\Gamma)}.$$

Use $p_0 = 1013$ hPa to estimate the pressure at $z = 5.0$ km and compare it with the isothermal scale-height estimate using $H = 8.2$ km.

Modelling connection.

The following problems connect vertical-profile physics to two useful modelling ideas: conservation of dry potential temperature during adiabatic motion, and the onset of rain by collision–coalescence. Students who also take EART22001 will meet related ideas in a single-column model, but no knowledge of that unit is assumed here. The detailed cloud microphysics is revisited in Week 9.

IN CLASS Question 10: Why potential temperature is a useful conserved variable

A dry parcel begins at $p_1 = 950$ hPa and $T_1 = 285$ K. It is displaced adiabatically to $p_2 = 800$ hPa.

1. Calculate its initial potential temperature using $\theta = T(1000/p)^{0.286}$.
2. Assuming θ is conserved, calculate the parcel temperature at 800 hPa.
3. Explain why a numerical atmospheric model may predict or conserve θ rather than temperature directly for dry adiabatic motion.

YOU SHOULD TRY Question 11: A first look at collision–coalescence and rain formation

Two cloud droplets have radii 10 μm and 25 μm. In the Stokes regime take terminal speed to scale as $v_t \propto r^2$.

1. By what factor does the larger droplet fall faster than the smaller one?
2. Explain how this differential fall speed can produce collisions.
3. If every collision led to coalescence, would all droplets remain the same size? Explain why a broadened droplet spectrum helps rain formation accelerate.